

Day 26 Problem Set

Geometry Summer Credit | Cumulative Synthesis Review

Name:	Date:	Period:
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Directions

Choose a strategy before calculating. This set prepares you for the Day 26 weekly assessment.

Part A: Circles and Coordinate Geometry

1.	Write the equation of a circle with center $(-4, 2)$ and radius 7.
2.	A circle has equation $(x - 6)^2 + (y + 1)^2 = 64$. Name the center and radius.
3.	Find the distance between $(-3, 5)$ and $(9, 10)$.
4.	Find the midpoint of $(-8, 6)$ and $(4, -2)$.
5.	A circle has center $(1, -2)$ and passes through $(7, 6)$. Find its radius.
6.	A circle has radius 10 and central angle 72 degrees. Find arc length.
7.	A circle has radius 8 and central angle 135 degrees. Find sector area.

Part B: Measurement and Scale

8. Find the area of a trapezoid with bases 16 and 28 and height 9.

9. A 20 by 14 rectangle has a 6 by 5 rectangle removed. Find the remaining area.

10. Find the surface area of a rectangular prism with length 9, width 5, and height 4.

11. Find the volume of a cone with radius 6 and height 9 in terms of π .

12. A sphere has radius 5. Find its surface area in terms of π .

13. Two similar figures have scale factor 4. The smaller area is 18 square units. Find the larger area.

14. Two similar solids have scale factor 2. The smaller volume is 55 cubic units. Find the larger volume.

Day 26 Problem Set

Continued | Modeling and Spiral Synthesis

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| 15. | A solid has mass 720 g and volume 180 cm^3 . Find its density. |
| 16. | A package can weigh no more than 2000 g. It has volume 260 cm^3 and density 7 g/cm^3 . Does it meet the weight constraint? |
| 17. | A wall is 32 ft by 9 ft. Paint covers 350 square feet per gallon. How many gallons are needed for one coat? |
| 18. | A ramp rises 4 ft over a horizontal distance of 16 ft. Find the angle the ramp makes with the ground to the nearest degree. |
| 19. | A right triangle has angle 38 degrees and adjacent side 15. Find the hypotenuse to the nearest tenth. |
| 20. | Challenge: A circle has equation $x^2 + y^2 - 10x + 4y - 7 = 0$. Complete the square to find its center and radius. |